

RCIM Model-Bank Reproduction Forward Paper-Faithful Grid-Search Closeout

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

- campaign name: track1_bidirectional_paper_faithful_grid_search_campaign_2026-05-04_12_26_30
- started at: 2026-05-14T00:44:05+02:00
- finished at: 2026-05-15T07:07:30+02:00
- closed direction: forward
- refreshed archive root: models/paper_reference/rcim_track1/forward
- benchmark report: doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md

Family Results

Family	Run Instance	Mean MAE	Mean RMSE	Mean MAPE %	Exported ONNX	Exported PKL
SVM	2026-05-14-11-22-01__track1_paper_faithful_grid_search_forward_svr_campaign_validation	0.195558	0.303488	75.0189	19	19
MLP	2026-05-14-11-24-54__track1_paper_faithful_grid_search_forward_mlp_campaign_validation	0.219477	0.302771	4319.59	19	19
RF	2026-05-14-11-41-07__track1_paper_faithful_grid_search_forward_rf_campaign_validation	0.0807811	0.161448	22.1391	19	19
DT	2026-05-14-13-03-07__track1_paper_faithful_grid_search_forward_dt_campaign_validation	0.0954926	0.186076	40.5591	19	19
ET	2026-05-14-13-04-54__track1_paper_faithful_grid_search_forward_et_campaign_validation	0.120815	0.252001	39.387	19	19
ERT	2026-05-14-13-06-36__track1_paper_faithful_grid_search_forward_ert_campaign_validation	0.0644413	0.151992	18.2706	19	19
GBM	2026-05-14-15-36-42__track1_paper_faithful_grid_search_forward_gbm_campaign_validation	0.0836404	0.164166	27.7299	19	19
HGBM	2026-05-14-17-04-55__track1_paper_faithful_grid_search_forward_hgbm_campaign_validation	0.106506	0.189713	38.1651	19	19
XGBM	2026-05-14-19-27-11__track1_paper_faithful_grid_search_forward_xgbm_campaign_validation	0.123318	0.193609	58.1335	19	19
LGBM	2026-05-14-19-30-29__track1_paper_faithful_grid_search_forward_lgbm_campaign_validation	0.148417	0.222652	64.7701	19	19
ELM	2026-05-15-07-06-18__track1_paper_faithful_grid_search_forward_elm_campaign_validation	0.206277	0.277969	99.7177	19	19

Benchmark Status

Table	Green	Yellow	Red	Total
Forward Table 2 - Amplitude MAE	19	25	66	110
Forward Table 3 - Amplitude RMSE	21	28	61	110
Forward Table 4 - Phase MAE	23	21	55	99
Forward Table 5 - Phase RMSE	23	32	44	99

Reference Archive Refresh

Family	Archived Targets	Source Runs	Archive Root
SVM	19	1	models/paper_reference/rcim_track1/forward/svm_reference_models
MLP	19	1	models/paper_reference/rcim_track1/forward/mlp_reference_models
RF	19	1	models/paper_reference/rcim_track1/forward/rf_reference_models
DT	19	1	models/paper_reference/rcim_track1/forward/dt_reference_models
ET	19	1	models/paper_reference/rcim_track1/forward/et_reference_models
ERT	19	1	models/paper_reference/rcim_track1/forward/ert_reference_models
GBM	19	1	models/paper_reference/rcim_track1/forward/gbm_reference_models
HGBM	19	1	models/paper_reference/rcim_track1/forward/hgbm_reference_models
XGBM	19	1	models/paper_reference/rcim_track1/forward/xgbm_reference_models
LGBM	19	1	models/paper_reference/rcim_track1/forward/lgbm_reference_models
ELM	19	1	models/paper_reference/rcim_track1/forward/elm_reference_models

Best Campaign Representative

- run: track1_paper_faithful_grid_search_forward_ert
- family: ERT
- mean MAE: 0.0644413
- mean RMSE: 0.151992

Notes

- The original paper-family order for Tables 2 - 5 remains unchanged.
- ELM is archived and benchmarked as an operational RCIM Model-Bank Reproduction family because the completed campaign includes it.
- Backward RCIM Model-Bank Reproduction paper-reference archives were not modified by this forward-only closeout.